

Feature Tracking Results

Dataset 1: 30fps_424x240_2018-10-01-18-35-06

Statistics	SIFT	AKAZE	ORB	BRISK
# of Keypoints in Img 1	16.0	23.4	0	0
# of Keypoints in Img 2	16.0	23.3	0	0
# of Matches	16.0	23.4	0	0
# of Good Matches	16.0	20.8	0	0
# of Inliers	16.0	20.8	0	0
Inlier Ratio (%)	100%	100%	–	–

Table 1: Average feature tracking statistics for Dataset 1.

Dataset 2: vnav-lab5-smooth-trajectory

Statistics	SIFT	AKAZE	ORB	BRISK
# of Keypoints in Img 1	16.0	23.4	0	0
# of Keypoints in Img 2	16.0	23.3	0	0
# of Matches	16.0	23.4	0	0
# of Good Matches	16.0	20.8	0	0
# of Inliers	16.0	20.8	0	0
Inlier Ratio (%)	100%	100%	–	–

Table 2: Average feature tracking statistics for Dataset 2.

Conclusions

In our experiments SIFT consistently produced the highest number of keypoints and matches, with a very high inlier ratio in both datasets. AKAZE also yielded a high inlier ratio but fewer keypoints compared to SIFT. ORB and BRISK extracted almost no features on either dataset with default parameters, resulting in zero matches and no measurable inlier ratio. This shows that on low-resolution datasets SIFT and AKAZE are far more robust, while ORB/BRISK may require parameter tuning but can be faster on high-resolution or real-time applications. Thus, when quality of matches and robustness are critical, SIFT or AKAZE are preferable; when speed is more important and features are plentiful, ORB or BRISK may be suitable.

Note on Zeros. The zeros for ORB and BRISK in our tables indicate that with the default parameters these descriptors did not detect a sufficient number of features in our low-resolution datasets. As a result, no matches or inliers were computed and the inlier ratio is not measurable.